Intermediate modular curves with infinitely many quintic points

Proof and computational certificates

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Abstract

We classify the intermediate modular curves with infinitely many closed points of degree exactly five over $\mathbb{Q}$. There are 43 positive strict subgroup-defined pairs, each admitting a rational function of degree five. We separate the short mathematical arguments from the exact computations they require. Complete case tables, source checks, and freshly replayed computational certificates accompany the proof.

1 Statement and conventions

For $\{\pm 1\} \subseteq \Delta \subseteq (\mathbb{Z}/N\mathbb{Z})^\times$, set

$$\Gamma_\Delta(N) = \left\{ \begin{pmatrix} a & b \\ c & d \end{pmatrix} \in \mathrm{SL}_2(\mathbb{Z}) : c \equiv 0 \pmod{N},\ a \bmod N \in \Delta \right\}.$$

The smooth projective modular curve attached to this subgroup is denoted $X_\Delta(N)$. We keep the pairs (N, Δ), rather than identifying curves that happen to be isomorphic. A pair is *strict* if both subgroup inclusions above are strict. A quintic point means a closed point with residue field of degree exactly five, not a point of degree at most five.

Theorem 1 (Classification)**.** *The strict intermediate curves with infinitely many quintic points are the 43 pairs in Table 2. Each admits a $\mathbb{Q}$-rational function of degree exactly five. Every other strict intermediate curve has only finitely many quintic points.*

Including endpoints, $X_1(N)$ is positive precisely for $N \in \{1, \dots, 25\} \setminus \{23\}$. The positive X_0 levels are

$$\{1, \dots, 45\} \cup \{47, \dots, 58\} \cup \{61, 63, 64, 65, 67, 68, 72, 73, 75, 80, 81, 91, 109, 121, 125\}.$$

The endpoint assertions are inputs to the strict classification. The X_1 assertion comes from Derickx–Sutherland [1]; the X_0 assertion uses the published classification [8, Theorem 1.10] together with the thirty-level proof in `../A_x0_quintic/quintic_points.pdf` (Theorem A of the paper). Its three routes use rank-zero lifting, quotient class sieves and local rigidity of reducible divisor families. Its 36 computational certificates are in `../A_x0_quintic/`.

For completeness, the X_1 endpoint statement concerns exact degree five. The cyclic torsion orders in [1, Theorem 1.1] are precisely $1 \le N \le 25$, $N \neq 23$. At genus zero, the rational cusp gives $X_1(N) \simeq \mathbb{P}^1$. At positive genus in this range, rank zero from [1, Theorem 4.1] makes rational points finite, so the infinitely many marked elliptic curves over quintic fields give genuine quintic points. Conversely, infinitely many quintic points give infinitely many marked elliptic curves; uniform boundedness and finite fibers of the j-map force a torsion group occurring infinitely often. Its cyclic subgroup order must lie in the same list. Finally, [1, Corollary 2.4] gives exact degree-five functions at the positive rank-zero endpoints as well.

Family of arguments	Strict pairs through level 191
Projection and spectral gonality	829
Rank-zero Jacobians	82
Positive-rank density and elliptic families	57
Degree-five function constructions	43

Table 1: The four main groups of proof assignments.

2 Three elementary reductions

Lemma 2 (A function produces quintic points)**.** *If a geometrically integral curve $C/\mathbb{Q}$ admits a degree-five map to $\mathbb{P}^1$, then it has infinitely many quintic points.*

Proof. Hilbert irreducibility supplies infinitely many rational fibers whose degree-five coordinate algebra is a field. After omitting branch values, these fibers are closed points of degree five. □

Lemma 3 (Projection)**.** *Let $C \to D$ be finite over $\mathbb{Q}$. If D has finitely many rational points and finitely many quintic points, then C has finitely many quintic points.*

Proof. The residue degree of the image divides five, hence is one or five. There are finitely many such images and each fiber has finite support. □

The rational-point hypothesis cannot be dropped. In applications it follows from $g(D) > 1$ and Faltings' theorem [10]. A negative X_0 target cannot have genus at most two, since its rational cusp and the low-genus construction below would give infinitely many quintic points. Forgetting maps are defined over $\mathbb{Q}$: for $M \mid N$, reduction of Δ_N into Δ_M permits $(E, P) \mapsto (E, (N/M)P)$. Same-level subgroup inclusions are special cases. Iteration starts only from proved negatives, so it creates no circular dependencies.

Lemma 4 (Rank zero)**.** *Suppose $C(\mathbb{Q}) \neq \varnothing$ and $J(C)(\mathbb{Q})$ has rank zero. If C has no rational function of degree five, it has finitely many quintic points.*

Proof. A rational base point identifies $\mathrm{Pic}^5(C)(\mathbb{Q})$ with the finite group $J(C)(\mathbb{Q})$. Let D be an irreducible effective quintic divisor. If $h^0(D) > 1$, a nonconstant rational function in $L(D)$ has a nonzero pole divisor bounded by D. Since D is one closed point with coefficient one, that pole divisor must equal D. This is a degree-five function, a contradiction. Otherwise D is the unique effective representative of its class. There are only finitely many classes. □

Remark 5. The rank-zero assumption is essential to this argument. It will not be used implicitly in the positive-rank cases.

3 Enumeration and rational models

Primitive rows (c, d) modulo N, modulo multiplication by Δ, parametrize $\Gamma_\Delta(N)\backslash \mathrm{SL}_2(\mathbb{Z})$. The scalar action is free: if $u(c, d) = (c, d)$, primitivity implies $u = 1$. Thus the index is

$$\mu = \frac{N^2}{|\Delta|} \prod_{\ell \mid N} (1 - \ell^{-2}).$$

Right multiplication by the standard generators gives permutations $(c, d) \mapsto (d, -c)$ and $(c, d) \mapsto (c, c + d)$. Fixed points of the order-two and order-three permutations count the elliptic points; cycles of the latter translation count cusps. Therefore

$$g = 1 + \frac{\mu}{12} - \frac{e_2}{4} - \frac{e_3}{3} - \frac{c_\infty}{2}.$$

Enumerating subgroups by successively adjoining every unit is complete: every subgroup has a finite generating set. Full element sets, not their orders or genera, are used as identifiers. A second computation with Sage's abstract unit groups and `GammaH` checks all 1388 pairs through level 191, of which 1011 are strict.

The rational structure deserves explicit attention. Rational Fourier expansions at the Tate cusp describe the model parametrizing μ_N-markings, not automatically the analytically identical point-marked model [9, proof of Lemma 6.1.6]. Quotienting an elliptic curve by its marked subgroup and dualizing the isogeny interchanges a μ_N-marking and a $\mathbb{Z}/N\mathbb{Z}$-marking. These inverse constructions are over $\mathbb{Q}$ and interchange diamond a with diamond a^{-1}. Since Δ is closed under inversion, they descend to a $\mathbb{Q}$-isomorphism of the two intermediate models. In particular all these curves have a rational Tate cusp after this identification.

For genus at least three, the full weight-two cusp space supplies canonical differentials. A degree-d relation has weight $2d$ and is certified by vanishing through $\lfloor 2d\mu/12 \rfloor$. Extra Fourier coefficients are checked but do not replace this bound. The quadratic product rank distinguishes the hyperelliptic canonical image from the nonhyperelliptic case. In the latter case the known canonical image is embedded, of degree $2g - 2$ and genus g. Exact containment in the proposed scheme, followed by equality of Hilbert polynomials, identifies the whole scheme: the quotient ideal sheaf has zero Hilbert polynomial and hence vanishes. Merely matching genera would not suffice.

Before a reduction argument, the displayed equations must pass a separate test at the chosen prime. Their special fiber is required to have the same Hilbert polynomial as the generic canonical curve, to be reduced and irreducible, to have normalization genus equal to its arithmetic genus, and to have a rational point. Here is why these tests suffice. The flat closure of the generic curve is contained in the displayed integral scheme. Its special fiber has the generic Hilbert polynomial. Equality with the displayed special-fiber Hilbert polynomial therefore identifies these two special fibers by the same quotient-sheaf argument as above. Equality of normalization and arithmetic genera makes the integral special curve normal, hence smooth over the perfect residue field. Its rational point forces the constant field to be the residue field, giving geometric integrality. The flat closure is consequently a smooth proper model near the special fiber. This is the model on which specialization is used; the condition $p \nmid N$ alone is not a check on chosen coordinates.

4 Positive cases

If $g \leq 2$ and $P \in C(\mathbb{Q})$, Riemann–Roch gives $h^0(5P) = 6 - g$ and $h^0(4P) = 5 - g$. Thus a rational function in $L(5P) \setminus L(4P)$ has pole divisor exactly $5P$, giving a degree-five map.

In genus three, the echelon pivot exponents of the Fourier basis are the Weierstrass gaps at the Tate cusp: a form of order a yields a differential of order $a - 1$. If five is not a gap, then $h^0(5P) > h^0(4P)$, giving a function with pole divisor $5P$.

In genus four, nonhyperellipticity makes the canonical bundle separate all length-two subschemes. Therefore $K - P$ is base-point-free, including at P, and has degree five. A general pair

of sections again gives the desired map. Nonhyperellipticity is checked by quadratic product rank nine rather than seven.

The remaining twelve new positive cases use explicitly verified degree-five pole divisors on certified models. The four published cases are exactly those in Orlić's Theorems 1.4–1.5 [3]. Fresh model generation and the degree-five witness checks have been replayed for all twelve new cases. The fourteen canonical low-genus cases have also been replayed; the thirteen genus-at-most-two cases use the uniform argument above. Replay evidence is indexed in `REAUDIT.md`.

Positive strict pairs. L: genus at most two; C: canonical low-genus argument; W: explicit witness; P: published pentagonality.

Table 2: Positive strict intermediate pairs.

N	Generators of Δ	g	Proof
13	$\langle -1, 3\rangle$	0	L
13	$\langle -1, 5\rangle$	0	L
15	$\langle -1, 4\rangle$	1	L
16	$\langle -1, 7\rangle$	0	L
17	$\langle -1, 2\rangle$	1	L
17	$\langle -1, 4\rangle$	1	L
19	$\langle -1, 7\rangle$	1	L
20	$\langle -1, 9\rangle$	1	L
21	$\langle -1, 4\rangle$	1	L
21	$\langle -1, 8\rangle$	3	C
24	$\langle -1, 5\rangle$	3	C
24	$\langle -1, 7\rangle$	3	C
24	$\langle -1, 11\rangle$	1	L
25	$\langle -1, 4\rangle$	0	L
25	$\langle -1, 7\rangle$	4	C
26	$\langle -1, 3\rangle$	4	C
26	$\langle -1, 5\rangle$	4	C
27	$\langle -1, 8\rangle$	1	L
28	$\langle -1, 3\rangle$	4	C
28	$\langle -1, 13\rangle$	4	C
29	$\langle -1, 4\rangle$	4	C
30	$\langle -1, 11\rangle$	5	W
31	$\langle -1, 2\rangle$	6	P
31	$\langle -1, 5\rangle$	6	P
32	$\langle -1, 7\rangle$	1	L
32	$\langle -1, 15\rangle$	5	W
33	$\langle -1, 2\rangle$	5	W
34	$\langle -1, 9\rangle$	5	W
35	$\langle -1, 4\rangle$	5	W
36	$\langle -1, 11\rangle$	3	C
37	$\langle -1, 3\rangle$	4	C
37	$\langle -1, 8\rangle$	4	C
39	$\langle -1, 4\rangle$	5	W
40	$\langle -1, 7\rangle$	7	W
40	$\langle -1, 9, 11\rangle$	5	W
41	$\langle -1, 2\rangle$	5	W
44	$\langle -1, 5\rangle$	8	P
45	$\langle -1, 4\rangle$	5	W
48	$\langle -1, 11\rangle$	5	W
49	$\langle -1, 6\rangle$	3	C
50	$\langle -1, 9\rangle$	4	C
64	$\langle -1, 7\rangle$	5	W
125	$\langle -1, 4\rangle$	16	P

5 Uniform negative arguments

Projection to the completed X_0 classification excludes every level $N \geq 192$. It excludes 807 strict pairs below this boundary. Seven more pairs are excluded by other forgetting maps. The spectral bound $\operatorname{gon}_{\mathbb{C}} C \geq 119\mu/12000$ [3, Theorem 3.16], together with Frey's necessary bound [1, Proposition 2.2] for quintic density, excludes another fifteen pairs: the implementation uses the strict integer inequality $119\mu > 120000$.

For rank-zero curves, it remains to exclude degree-five functions. The published classifications of hyperelliptic, trigonal, tetragonal and pentagonal intermediate curves do this whenever $\operatorname{gon}_{\mathbb{Q}} C \geq 6$. The subgroup lists are expanded as element sets before comparison; genus or group order alone is not a sufficient identifier. Nonvanishing central values establish rank zero by the modular nonvanishing theorem. For the nontrivial-nebentypus factors occurring here, we use Kato's Corollary 14.3, as applied to modular Jacobians in Derickx–Sutherland, Section 4 [2, 1]; citing only the trivial-nebentypus version would not justify all intermediate cases. The Magma overload used here factors the modular abelian variety and tests exact modular-symbol L-ratios. It is not a numerical approximation to $L(1)$. Rank zero of $J_1(N)$ also descends to its intermediate quotients up to isogeny.

Two rank-zero exceptions, of genera seven and nine at levels 35 and 55, are geometrically bielliptic [6, Theorem 0.2 and Table 6]. A degree-two elliptic map and a degree-five rational map generate the function field, since their degrees are coprime. Castelnuovo–Severi would give $g \leq 2+(2-1)(5-1) = 6$. Thus no quintic function exists even geometrically. A tetragonal curve is not excluded just because its gonality happens to equal four.

6 Exhaustive rational divisor-class sieves

Let C have good reduction at an odd prime p and a rational point P. Suppose rank zero has been proved and an explicit subgroup of rational divisor classes is proved to exhaust $J(C)(\mathbb{Q})$. Reduction on rational torsion over the unramified field $\mathbb{Q}_p$ is injective. Enumerate every effective degree-five divisor on $C_{\mathbb{F}_p}$, using all closed places of degrees one through five and all partitions of five. Keep the classes with $h^0 \geq 2$ that belong to the known rational subgroup, shifted by $5P$. This includes the specialization of every rational quintic pencil, by upper semicontinuity. Over a finite field, an effective rational line-bundle class has a rational effective representative, so enumeration of rational effective divisors omits no relevant class.

Lift each surviving class to a rational divisor and compute its full Riemann–Roch space. A bounded search may find a convenient representative, but every surviving class must be checked. Injectivity of reduction identifies the lift with the intended rational class. If its section space has dimension at most one, it is not a pencil. If the dimension is two and the complete system has a base point, it cannot give a degree-five map. These checks exclude every rational quintic function; the rank-zero lemma then excludes quintic density.

The assertion that the rational subgroup is complete is crucial. At level 40, a full integral Manin-symbol lattice, intersected with the cusp-boundary kernel, gives absolute integral homology. On its rational quotient by this lattice, rational torsion is fixed by complex conjugation and annihilated, away from the residue characteristic, by

$$1 - T_p + p\langle p\rangle = 1 - T_p + T_p^2 - T_{p^2}.$$

Smith normal forms give rigorous upper bounds, with each residue prime omitted when bounding its own primary part. Their combination with Frobenius-order bounds gives 4032. Explicit

rational point differences generate 4032 classes on reduction, proving equality. Twelve special classes remain, and all twelve are checked over $\mathbb{Q}$.

For the two level-48 cases, use the published description of $J_1(48)(\mathbb{Q})_{\mathrm{tors}}$ by rational cusp-orbit divisors [7, Theorem 4.13]. Their embedding is recomputed in full integral homology. Transfer for the degree-four forgetting map is obtained by summing lifted paths over all sheets; it induces pullback on Jacobians. Every rational torsion class on the target must transfer into the computed rational cuspidal group upstairs. This restricts the target Hecke bounds to 2304 and 12800, respectively. Explicit closed-cusp classes attain both bounds. The two finite-field sieves then leave four and 24 classes, all checked rationally. Pullback need not be injective: taking its inverse image is the correct restriction.

Characterwise rational decompositions are used only for isogeny invariants, not as integral homology lattices. For a rational-point divisor D, the final Riemann–Roch computations can instead use canonical hyperplanes: the saturated ideal $I_C + I(P)^m$ cuts out mP on the smooth curve. Its linear part is $H^0(K-mP)$. Intersect these spaces and apply Riemann–Roch. Equalities $h^0(D-P) = h^0(D)$ certify base points.

7 A formal lifting lemma

The same local argument underlies the two remaining computational methods. It is stronger than a dimension count on the special fiber.

Lemma 6. *Let R be a complete discrete valuation ring. Let $\mathcal{P}/R$ be smooth of relative dimension g, and $\mathcal{T} \subseteq \mathcal{P}$ a closed subscheme. Suppose a smooth formal r-dimensional branch through a residue point maps into $\mathcal{T}$ and is a formal immersion into $\mathcal{P}$. If the relative tangent space of $\mathcal{T}$ at that residue point has dimension at most r, then the completion of $\mathcal{T}$ there is exactly that branch.*

Proof. Write the completed ambient ring as $R[[x_1, \dots, x_g]]$. The tangent bound supplies $g - r$ equations in the ideal of $\mathcal{T}$ with independent transverse linear terms modulo the maximal ideal of R. The formal implicit function theorem eliminates these variables and leaves $R[[t_1, \dots, t_r]]$. The contained branch has the same tangent dimension, so its map into this ring is an isomorphism by the formal inverse function theorem. Every remaining equation vanishes on that branch and hence is zero. In particular, no additional vertical or infinitesimal components survive. □

7.1 Petri lifting for rank-zero curves

Suppose explicit base-point-free quartic bundles A_i have $h^0 = 2$ over $\mathbb{Q}$ and after good reduction. Suppose an exhaustive computation shows that every class in $W^1_5(\mathbb{F}_p)$ is some $A_i + P$, with $P \in C(\mathbb{F}_p)$, has exactly two sections, and has Petri tangent dimension one. Each relative Abel curve $A_i + C$ lies in relative W^1_5. Its differential is injective. Lemma 6 identifies the whole local W^1_5 with this known branch.

A hypothetical rational quintic pencil would therefore lie over $\mathbb{Q}_p$ on some $A_i + C$. Upper semicontinuity bounds its number of sections by two. Both sections then vanish at the added point, contradicting base-point-freeness. The rank-zero lemma gives finiteness. The tangent calculation is the exact Petri formula

$$\dim T_{[D]} W^1_5 = g - \operatorname{rank}\big(H^0(D) \otimes H^0(K-D) \longrightarrow H^0(K)\big) \quad (h^0(D) = 2).$$

8 Positive rank: density theorems and elliptic maps

Here it is necessary to exclude infinite divisor families, not merely moving linear systems. First check from the lower-degree classifications that points of each degree one through four are finite. Quintic density would then have minimum density degree five. Kadets–Vogt's genus bound [4, Theorem 1.3] forces a degree-five map to $\mathbb{P}^1$ or a positive-rank elliptic curve when $g \geq 12$. Their arithmetic classification for $g \geq 9$ additionally allows a Debarre–Fahlaoui normalization [4, Proposition 6.3, arithmetic row]. All 51 uses of these density theorems have been compared, by full subgroup element sets, with the lower-degree lists in [5, Theorems 1.4–1.6]; the executable `check_density_hypotheses.py` records that check.

The latter alternative lies on $\mathrm{Sym}^2 E$ in numerical class $B = (5+m)H - mF$, $1 \leq m \leq 5$, with $H^2 = H \cdot F = 1$, $F^2 = 0$, and $K = -2H + F$. Direct adjunction gives

$$p_a(B) = 1 + \tfrac{1}{2}(B^2 + B \cdot K) = 11 - \tfrac{1}{2}m(m-1).$$

If the normalization has genus at least nine, then $m = 1$ or 2. The addition map to E consequently has degree $B \cdot F = 6$ or 7. Thus excluding elliptic maps of degrees five, six and seven handles these alternatives as well. The arithmetic construction, over $\mathbb{Q}$, is needed here; the geometric classification alone would not suffice.

At the levels where the optimal-parametrization theorem applies, every positive-rank elliptic factor has full conductor N, multiplicity one, and odd analytic rank. For $N \leq 800$ the composition through $X_0(N)$ is optimal [5, Corollary 4.10]. If $t = [(\mathbb{Z}/N\mathbb{Z})^\times : \Delta]$ and d is the optimal modular degree on $X_0(N)$, every elliptic map in that isogeny class has degree a multiple of td. In the applications $t, d > 1$, so degree five is impossible. At levels 37 and 57 the sharper bounds are 12 and eight. At 73, 109 and 125 the relevant factor computations exclude positive-rank elliptic quotients. Every application also needs its separate exclusion of rational quintic functions.

9 Positive rank: the remaining elliptic families

Suppose the only potentially positive-rank simple factor of $J(C)$ is one copy of an elliptic curve. Let $A \subseteq J(C)$ be that elliptic subvariety. The rational Mordell–Weil group has Zariski closure contained in finitely many translates of A. An infinite intersection of W_5 with one such translate is dense in that elliptic curve, so the whole translate lies in W_5.

At level 43, good reduction at two gives $|A(\mathbb{F}_2)| = 5$ and $J(\mathbb{F}_2) \simeq \mathbb{Z}/38 \oplus \mathbb{Z}/1330$. There is a unique order-five subgroup. Exhaustive enumeration gives $|W_5(\mathbb{F}_2)| = 2133$ and no full coset of that subgroup in W_5. Prime-to-two torsion specializes injectively, so no rational elliptic translate can exist. Thus $W_5(\mathbb{Q})$ is finite.

For levels 53, 61 and the three exceptional level-65 curves, there is a degree-four map $f : C \to E$ whose pullback image is A. The surface

$$Z = \{f^*Q + P : Q \in \mathrm{Pic}^1(E),\ P \in C\} \subseteq \mathrm{Pic}^5(C)$$

is an unavoidable family of reducible effective divisors. At a good odd prime p, let $B = A(\mathbb{F}_p)$. The decisive finite-field tests are:

1. every full B-coset in $W_5(\mathbb{F}_p)$ is a known fiber-plus-point coset;

2. the intersection of the corresponding translated tangent spaces has dimension at most two;

3. $A \times C \to \mathrm{Pic}^5(C)$ has rank-two differential at every relevant rational reduction preimage.

Since $p \nmid 4$, $f_* f^* = [4]$ identifies a smooth elliptic subgroup over $\mathbb{Z}_p$. Lift each $b \in B$ to some $a_b \in A(\mathbb{Z}_p)$ and intersect the relative loci $W_5 - a_b$. Neither torsion lifts nor a subgroup of lifts are required. Both Z and every hypothetical A-translate in W_5 are contained in this intersection. The tests and Lemma 6 identify its completion at each relevant point with the known smooth two-dimensional branch. Consequently each such elliptic translate is contained in the closed surface Z.

An irreducible quintic divisor cannot occur on Z when no quintic function exists. Indeed, its h^0 must be one, so its unique effective representative is $f^*Q + P$. The leftover point P is unique: distinct fibers are disjoint, whereas removing two points from a degree-five divisor leaves common support of degree at least three. Galois invariance therefore makes P rational, contradicting irreducibility. Only finitely many isolated rational classes remain outside these families, and each contributes at most one irreducible divisor.

The tangent computations use a consistent differential basis. At a class with $h^0(D) = 1$, the annihilator of $T_{[D]}W_d$ is $H^0(K - D)$. Ignoring classes with more sections only weakens the upper bound. Intersections of tangents at known quartic fibers identify the line T_0A; comparison with Abel tangents checks the final immersion.

The numerical checks for these five curves are summarized below. The last column counts full B-cosets in $W_5(\mathbb{F}_p)$, all of which are known fiber-plus-point cosets. Every one has tangent-intersection dimension two and passes the immersion test.

N	Generators of Δ	g	p	$\lvert B\rvert$	Full cosets
53	$\langle -1, 4\rangle$	8	5	6	8
61	$\langle -1, 4\rangle$	8	5	9	10
65	$\langle -1, 2, 7\rangle$	9	7	12	4
65	$\langle -1, 3\rangle$	11	3	6	8
65	$\langle -1, 6\rangle$	11	3	6	4

10 Exceptional case index

Table 3 specifies the curves requiring the three divisor-class methods and the elliptic-family method. Full subgroup element sets and certificate paths are in `all_cases.csv`. The executable `replay_special.py` assigns each of these twelve cases a separate working copy, re-executes its certificate, and retains the resulting evidence. It reuses the supplied canonical-model data; that distinction from fresh model generation is intentional.

11 Conclusion and verification scope

Proof of the classification theorem. The completed X_0 classification and the projection lemma exclude all $N \geq 192$. The independent subgroup census lists every remaining pair. For the 43 strict pairs in Table 2, the indicated construction gives an exact degree-five function, hence infinitely many quintic points. Table 4 assigns each of 154 strict pairs one of the direct negative arguments proved above. The remaining negative pairs are the 807 projections to negative X_0 curves and the seven explicit other forgetting maps in the appendix. Every projection target has genus greater than one and an already established negative result. These disjoint assignments account for all $43 + 154 + 807 + 7 = 1011$ strict pairs. The endpoint arguments at the start complete the classification. □

Table 3: The twelve exceptional negative certificates.

N	Generators of Δ	g	Method
36	$\langle -1, 17\rangle$	7	Petri lifting
39	$\langle -1, 5\rangle$	9	Petri lifting
40	$\langle -1, 3\rangle$	7	Rational classes
43	$\langle -1, 2\rangle$	9	Empty cosets
48	$\langle -1, 5\rangle$	7	Rational classes
48	$\langle -1, 7, 17\rangle$	7	Rational classes
53	$\langle -1, 4\rangle$	8	Elliptic families
61	$\langle -1, 4\rangle$	8	Elliptic families
65	$\langle -1, 2, 7\rangle$	9	Elliptic families
65	$\langle -1, 3\rangle$	11	Elliptic families
65	$\langle -1, 6\rangle$	11	Elliptic families
75	$\langle -1, 4\rangle$	9	Petri lifting

The code checks the stated finite-field, divisor-class, rank and model identities; the deductions from these identities are the mathematical arguments above. This is not a formal proof-assistant verification. `SOURCE_AUDIT.md` records the published inputs and their hypotheses. `REAUDIT.md` records the completed mathematical review, fresh computations and portability tests. The command `python3 verify.py` checks the sealed package and its complete case assignments without requiring the working repository. The machine-readable `all_cases.csv` retains the complete subgroup, answer, projection target and proof metadata for every enumerated pair. It is an index to the evidence, not a substitute for the arguments.

A Complete direct negative assignments

The table below accounts for all 154 direct negative assignments. Together with the 807 projections to negative X_0 curves and seven other forgetting maps, these give $154 + 807 + 7 = 968$ negative strict pairs. The 43 positive pairs complete the 1011 strict pairs through level 191; projection to X_0 excludes all larger levels. The CSV records the exact target for each forgetting map and the certificate or published input for every row.

For convenience, the seven non-X_0 projections are displayed explicitly:

N	Source generators	M	Target generators
63	$\langle -1, 5\rangle$	63	$\langle -1, 4, 5\rangle$
68	$\langle -1, 13\rangle$	34	$\langle -1, 13\rangle$
68	$\langle -1, 33\rangle$	34	$\langle -1, 13\rangle$
80	$\langle -1, 3\rangle$	40	$\langle -1, 3\rangle$
80	$\langle -1, 9, 31\rangle$	40	$\langle -1, 3\rangle$
80	$\langle -1, 11\rangle$	40	$\langle -1, 11\rangle$
80	$\langle -1, 19\rangle$	40	$\langle -1, 19\rangle$

The proof codes refer to the preceding arguments:

Code	Argument
S	Spectral gonality bound $119\mu > 120000$
Z	Rank zero and published rational gonality at least six
B	Rank zero and the bielliptic double-cover inequality
R	Complete rational divisor-class sieve
T	Rank zero and Petri lifting
E	Minimum density five and elliptic-degree obstruction
E0	Minimum density five and no positive-rank elliptic factor
D	Debarre–Fahlaoui obstruction in genus nine or ten
C	No elliptic translate survives reduction
F	Surviving elliptic families contain only reducible divisors

Table 4: The 154 strict pairs with a direct negative argument: 139 curves with an individual argument and the 15 curves of large index (code S, $119\mu > 120000$). Projections are listed separately.

N	Generators of Δ	g	μ	Proof
29	$\langle -1, 12\rangle$	8	210	Z
33	$\langle -1, 10\rangle$	11	240	Z
34	$\langle -1, 13\rangle$	9	216	Z
35	$\langle -1, 8\rangle$	7	144	B
35	$\langle -1, 11\rangle$	9	192	Z
35	$\langle -1, 6\rangle$	13	288	Z
36	$\langle -1, 17\rangle$	7	216	T
37	$\langle -1, 10\rangle$	10	228	D
37	$\langle -1, 6\rangle$	16	342	E
38	$\langle -1, 7\rangle$	10	180	Z
39	$\langle -1, 5\rangle$	9	168	T
39	$\langle -1, 16\rangle$	9	224	Z
39	$\langle -1, 14\rangle$	17	336	Z
40	$\langle -1, 3\rangle$	7	144	R
40	$\langle -1, 9\rangle$	13	288	Z
40	$\langle -1, 11\rangle$	13	288	Z
40	$\langle -1, 19\rangle$	9	288	Z
41	$\langle -1, 4\rangle$	11	168	Z
41	$\langle -1, 3\rangle$	11	210	Z
41	$\langle -1, 9\rangle$	21	420	Z
42	$\langle -1, 5\rangle$	9	192	Z
42	$\langle -1, 13\rangle$	13	288	Z
43	$\langle -1, 2\rangle$	9	132	C
43	$\langle -1, 6\rangle$	15	308	E
44	$\langle -1, 21\rangle$	16	360	Z
45	$\langle -1, 8\rangle$	11	216	Z
45	$\langle -1, 14\rangle$	9	288	Z
45	$\langle -1, 19\rangle$	21	432	Z
48	$\langle -1, 5\rangle$	7	192	R
48	$\langle -1, 7, 17\rangle$	7	192	R
48	$\langle -1, 7\rangle$	19	384	Z
48	$\langle -1, 17\rangle$	19	384	Z
48	$\langle -1, 23\rangle$	13	384	Z
49	$\langle -1, 18\rangle$	19	392	Z

N	Generators of Δ	g	μ	Proof
50	$\langle -1, 7\rangle$	22	450	Z
51	$\langle -1, 2\rangle$	9	144	Z
51	$\langle -1, 4\rangle$	17	288	Z
51	$\langle -1, 16\rangle$	33	576	Z
52	$\langle -1, 3\rangle$	9	168	Z
52	$\langle -1, 5\rangle$	13	252	Z
52	$\langle -1, 9\rangle$	19	336	Z
52	$\langle -1, 25\rangle$	25	504	Z
53	$\langle -1, 4\rangle$	8	108	F
53	$\langle -1, 23\rangle$	40	702	E
54	$\langle -1, 17\rangle$	10	324	Z
55	$\langle -1, 4\rangle$	9	144	B
55	$\langle -1, 16\rangle$	17	288	Z
55	$\langle -1, 12\rangle$	21	360	Z
55	$\langle -1, 21\rangle$	41	720	Z
56	$\langle -1, 3\rangle$	9	192	Z
56	$\langle -1, 5\rangle$	11	192	Z
56	$\langle -1, 17\rangle$	11	192	Z
56	$\langle -1, 13, 15\rangle$	13	288	Z
56	$\langle -1, 9\rangle$	21	384	Z
56	$\langle -1, 13\rangle$	31	576	Z
56	$\langle -1, 15\rangle$	31	576	Z
56	$\langle -1, 27\rangle$	25	576	Z
57	$\langle -1, 2\rangle$	9	160	D
57	$\langle -1, 11\rangle$	13	240	E
57	$\langle -1, 7\rangle$	25	480	E
57	$\langle -1, 20\rangle$	43	720	E
58	$\langle -1, 5\rangle$	12	180	E
58	$\langle -1, 17\rangle$	36	630	E
61	$\langle -1, 4\rangle$	8	124	F
61	$\langle -1, 8\rangle$	12	186	E
61	$\langle -1, 21\rangle$	16	310	E
61	$\langle -1, 3\rangle$	26	372	E
61	$\langle -1, 13\rangle$	36	620	E
61	$\langle -1, 11\rangle$	56	930	E
63	$\langle -1, 4, 5\rangle$	9	192	Z
63	$\langle -1, 2\rangle$	17	288	Z
63	$\langle -1, 11\rangle$	17	288	Z
63	$\langle -1, 10\rangle$	17	288	Z
63	$\langle -1, 13\rangle$	13	288	Z
63	$\langle -1, 4\rangle$	33	576	Z
63	$\langle -1, 17\rangle$	33	576	Z
63	$\langle -1, 20\rangle$	25	576	Z
63	$\langle -1, 8\rangle$	49	864	Z
64	$\langle -1, 15\rangle$	13	384	Z
64	$\langle -1, 31\rangle$	37	768	Z
65	$\langle -1, 2, 7\rangle$	9	168	F
65	$\langle -1, 3\rangle$	11	168	F
65	$\langle -1, 6\rangle$	11	168	F
65	$\langle -1, 8, 12\rangle$	13	252	E
65	$\langle -1, 2\rangle$	19	336	E

N	Generators of Δ	g	μ	Proof
65	$\langle -1, 7\rangle$	19	336	E
65	$\langle -1, 9\rangle$	21	336	E
65	$\langle -1, 8, 14\rangle$	25	504	E
65	$\langle -1, 12\rangle$	31	504	E
65	$\langle -1, 21\rangle$	31	504	E
65	$\langle -1, 4\rangle$	41	672	E
65	$\langle -1, 8\rangle$	55	1008	E
65	$\langle -1, 14\rangle$	61	1008	E
65	$\langle -1, 18\rangle$	55	1008	E
67	$\langle -1, 3\rangle$	15	204	E
67	$\langle -1, 29\rangle$	45	748	E
68	$\langle -1, 9\rangle$	13	216	Z
72	$\langle -1, 5\rangle$	13	288	Z
72	$\langle -1, 7\rangle$	13	288	Z
72	$\langle -1, 11\rangle$	9	288	Z
72	$\langle -1, 17, 19\rangle$	21	432	Z
72	$\langle -1, 23\rangle$	25	576	Z
72	$\langle -1, 17\rangle$	49	864	Z
72	$\langle -1, 19\rangle$	49	864	Z
72	$\langle -1, 35\rangle$	41	864	Z
73	$\langle -1, 6\rangle$	9	148	D
73	$\langle -1, 7\rangle$	13	222	E
73	$\langle -1, 2\rangle$	19	296	E
73	$\langle -1, 3\rangle$	25	444	E
73	$\langle -1, 10\rangle$	43	666	E
73	$\langle -1, 8\rangle$	55	888	E
73	$\langle -1, 27\rangle$	85	1332	S
75	$\langle -1, 4\rangle$	9	240	T
75	$\langle -1, 14\rangle$	17	480	Z
75	$\langle -1, 7\rangle$	37	600	Z
75	$\langle -1, 26\rangle$	73	1200	S
80	$\langle -1, 3, 13\rangle$	15	288	Z
80	$\langle -1, 7, 9\rangle$	15	288	Z
80	$\langle -1, 9, 11\rangle$	13	288	Z
80	$\langle -1, 7\rangle$	35	576	Z
80	$\langle -1, 13\rangle$	35	576	Z
80	$\langle -1, 17\rangle$	35	576	Z
80	$\langle -1, 9\rangle$	69	1152	S
80	$\langle -1, 31\rangle$	69	1152	S
80	$\langle -1, 39\rangle$	57	1152	S
81	$\langle -1, 8\rangle$	10	324	Z
81	$\langle -1, 26\rangle$	46	972	Z
91	$\langle -1, 3, 4\rangle$	13	224	E
91	$\langle -1, 2\rangle$	21	336	E
91	$\langle -1, 11\rangle$	21	336	E
91	$\langle -1, 5\rangle$	23	336	E
91	$\langle -1, 6\rangle$	23	336	E
91	$\langle -1, 9, 12\rangle$	25	448	E
91	$\langle -1, 3\rangle$	41	672	E
91	$\langle -1, 4\rangle$	41	672	E
91	$\langle -1, 25\rangle$	45	672	E

N	Generators of Δ	g	μ	Proof
91	$\langle -1, 36\rangle$	45	672	E
91	$\langle -1, 8\rangle$	67	1008	E
91	$\langle -1, 9\rangle$	81	1344	S
91	$\langle -1, 12\rangle$	89	1344	S
91	$\langle -1, 16\rangle$	81	1344	S
91	$\langle -1, 22\rangle$	89	1344	S
91	$\langle -1, 27\rangle$	133	2016	S
109	$\langle -1, 3\rangle$	16	220	E0
109	$\langle -1, 2\rangle$	22	330	E0
109	$\langle -1, 4\rangle$	46	660	E0
109	$\langle -1, 8\rangle$	64	990	E0
109	$\langle -1, 45\rangle$	136	1980	S
109	$\langle -1, 33\rangle$	208	2970	S
121	$\langle -1, 10\rangle$	26	660	E
121	$\langle -1, 3\rangle$	106	1452	S
125	$\langle -1, 7\rangle$	36	750	E0
125	$\langle -1, 24\rangle$	76	1500	S
125	$\langle -1, 57\rangle$	256	3750	S

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